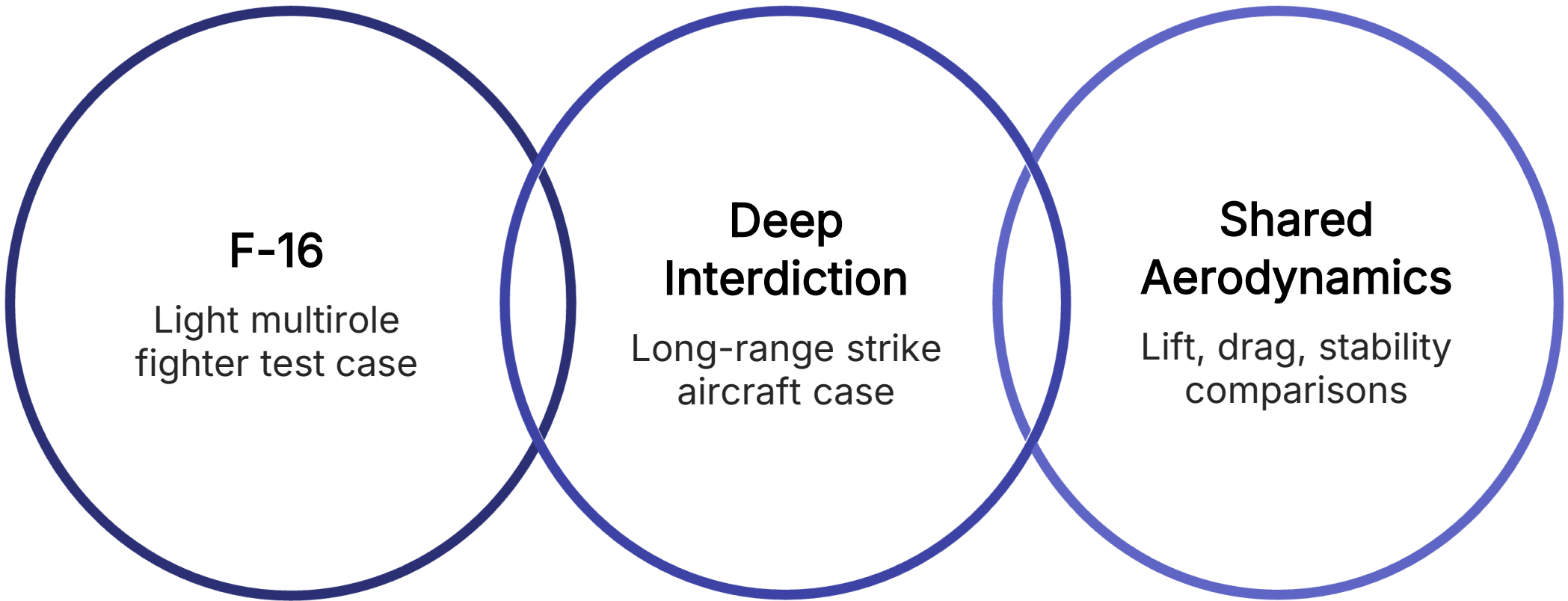


Code Validation: Theoretical vs. Simulated Aerodynamics

Our validation process systematically compares theoretical "expected" aerodynamic values with results generated by our simulation code. The evaluation focuses on critical performance metrics across different flight regimes and aircraft designs.



Aerodynamic Calculation Methodology

The formulae were taken from *Fundamentals of Aircraft and Airship Design by Nicolai and Carichner*. The plots in the book were used to interpolate data for the desired parameter.

Key Aerodynamic Parameters:

Wave Drag Coefficient (C_{dw})

- Calculated using supersonic area rule and Laitone's rule
- $C_{dw} = f(\text{Mach, aircraft geometry, thickness ratio})$
- Critical for supersonic flight regimes ($M > 1.0$)

Zero-Lift Drag (C_{d0})

- $C_{d0} = C_{d0_wing} + C_{d0_body}$
- Comprises skin friction drag and wave drag components
- $C_{d0_body} = C_{d0_body}(\text{skin}) + C_{d0_body}(\text{wave})$

Lift-to-Drag Ratio (L/D)

- $L/D = CL / CD$
- $CL = \text{dynamic pressure} \times \text{wing area} \times \text{lift coefficient}$
- $CD = C_{dw} + C_{d0} + C_{Di}$ (wave + zero-lift + induced drag)

Oswald Efficiency Factor (e)

- $e = 1 / (1 + \delta \times \pi \times AR)$
- Accounts for induced drag effects
- Higher e indicates better aerodynamic efficiency
- Typical range: 0.6-0.9 for fighter aircraft

Induced Drag Coefficient

- $C_{Di} = (CL^2) / (\pi \times e \times AR)$
- $AR = \text{aspect ratio}$
- Directly related to wing planform efficiency

F-16 Input Specifications & Geometry

Mass & Flight Conditions

- Mass: 16,875 kg (MTOW) / 9,682.5 kg (Maneuver Weight)
- Mach Numbers: 1.05, 1.5, 2.0 (Supersonic Regimes)

Airfoil Characteristics

- Type: NACA 64A204 (Biconvex Assumption)
- Thickness-to-Chord (t/c): 4%

Wing Geometry

- Wing Span: 9.144 m
- Aspect Ratio: 3
- Leading Edge Sweep: 40°

Fuselage Configuration

- Centerbody: 11.89 m
- Nose: Conical (20° semi-vertex angle)
- Aftbody: Truncated

F-16 Key Results & Code Accuracy

Parameter	Expected Value	Code Output	Difference	Percentage difference
M @ 1.05				
Cdw@1.05	0.0261	0.0231	-0.003	11.49%
Cd0@1.05	0.0444	Wing-0.0323, Body-0.0105, Total= <u>0.0428</u>	-0.0016	-3.60%
L/D@1.05	Not known	3.612	-	-
M @ 1.5				
Cdw@1.5	0.0213	0.022	0.0007	3.28%
Cd0@1.5	0.0448	Wing-0.0308, Body-0.0100, Total= <u>0.0408</u>	-0.004	-8.92%
L/D@1.5	Not known	2.793	-	-
M @ 2				
Cdw@2	0.0189	0.0205	0.0016	8.46%
Cd0@2	0.0458	Wing-0.0289, Body-0.0096, Total= <u>0.03857</u>	0.00723	-15.20%
L/D@2	Not known	1.887	-	-
e(Oswald efficiency)	0.703	0.756	0.053	7.54%
Cl alpha wing	0.056	0.0495	0.0062	11.01%

The results show reasonable agreement, with percentage differences within acceptable ranges for initial validation.

F-16 Detailed Analysis & Component Breakdown

Drag Component Analysis at Different Mach Numbers:

1	2	3
At Mach 1.05 Wave Drag (Cdw): 0.0231 (Code) vs 0.0261 (Expected) → -11.49% underestimation Zero-Lift Drag (Cd0): 0.0428 (Code breakdown: Wing-0.0323, Body-0.0105) - The underestimation at transonic speeds suggests the code may not fully capture shock formation effects	At Mach 1.5 Wave Drag (Cdw): 0.0220 (Code) vs 0.0213 (Expected) → +3.28% overestimation Zero-Lift Drag (Cd0): 0.0408 (Code breakdown: Wing-0.0308, Body-0.0100) - Excellent agreement at mid-supersonic speeds indicates robust model performance	At Mach 2.0 Wave Drag (Cdw): 0.0205 (Code) vs 0.0189 (Expected) → +8.46% overestimation Zero-Lift Drag (Cd0): 0.03857 (Code breakdown: Wing-0.0289, Body-0.0096) - Higher Mach number shows increased drag prediction, possibly due to compressibility effects

Key Observations:

- Oswald Efficiency:** 0.756 (Code) vs 0.703 (Expected) → +7.54% improvement
- Cl alpha wing:** 0.0495 (Code) vs 0.056 (Expected) → -11.01% underestimation
- The code shows better efficiency prediction but slightly underestimates lift slope

F-16 L/D Ratios and Trend Observation

Lift-to-Drag Performance

- At Mach 1.05, the code calculated an L/D ratio of **3.612**.
- At Mach 2.0, the L/D ratio decreased to **1.887**, reflecting increased drag at higher supersonic speeds.

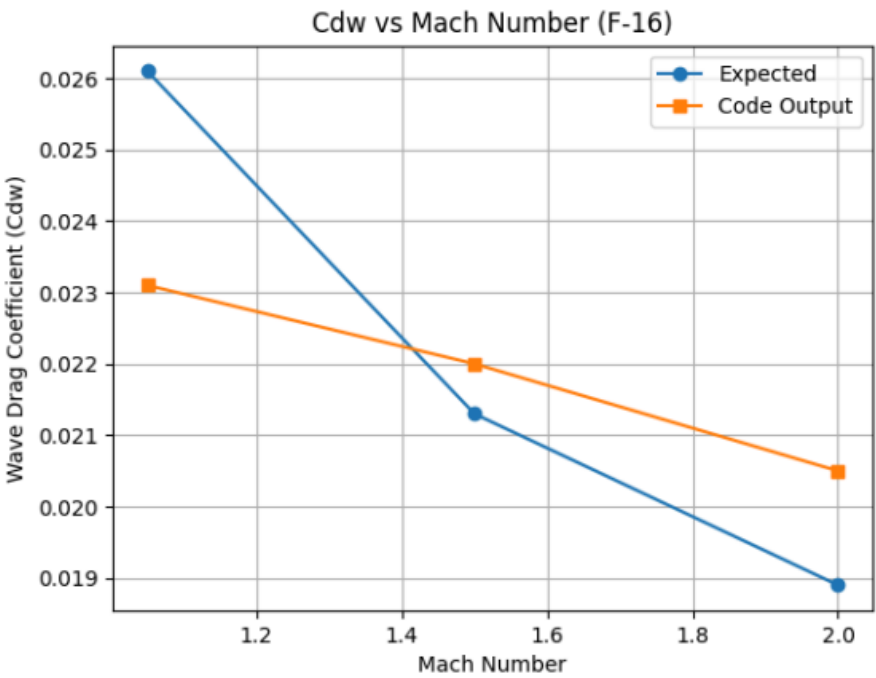
The decreasing L/D ratio with increasing Mach number is a critical characteristic of supersonic flight, indicative of the escalating wave drag and other compressible flow effects.

Wave Drag Trend Analysis

The parity plot reveals a consistent trend:

- Lower Supersonic Speeds (Mach 1.05):** The code slightly underestimates wave drag.
- Higher Supersonic Speeds (Mach 2.0):** The code tends to overestimate wave drag.

This observation highlights specific speed regimes where further model refinement might be beneficial.



Mach Number Effects on Aerodynamic Efficiency

Observations on Lift-to-Drag (L/D) Ratio:

- At Mach 1.05: L/D = 3.612 (highest efficiency)
- At Mach 1.5: L/D = 2.793 (intermediate)
- At Mach 2.0: L/D = 1.887 (lowest efficiency)

The 48% reduction in L/D from Mach 1.05 to Mach 2.0 demonstrates the severe penalty of supersonic flight on aerodynamic efficiency. This is primarily due to:

- Exponential increase in wave drag with Mach number
- Shock-induced pressure changes reducing lift effectiveness
- Compressibility effects on boundary layer behavior

The code successfully captures this trend, validating its ability to model transonic and supersonic aerodynamic phenomena across a wide speed range.

Deep Interdiction Aircraft: Input Specifications

General Parameters

- MTOW: 51,739 kg
- Evaluation Mach: 1.6

Wing Geometry

- Span: 13 m
- Aspect Ratio: 1.27
- Tip Chord: 0 m (Delta-style wing)

Fuselage Configuration

- Nose Length: 12.5 m
- Aftbody Length: 13 m
- Maximum Diameter: 3.3 m

Nose Type

- Shape: Conical
- Semi-Vertex Angle: 45°

Performance Comparison at Mach 1.6

Parameter	Expected Result	Code Output	Difference	% Error
Cdw	0.01624	0.01484	-0.0014	-8.62%
Cd0	0.0205	0.0224	0.0019	9.27%
Cl	0.146	0.1365	-0.0095	-6.51%
Cd	0.0315	0.0348	0.0033	10.48%
L/D	4.65	3.917	-0.733	-15.76%

The comparison reveals some notable differences between expected and code outputs for the Deep Interdiction Aircraft's performance at Mach 1.6. Specifically, the Lift-to-Drag (L/D) ratio shows the largest percentage error, indicating a significant deviation in overall aerodynamic efficiency. The drag coefficient (Cd) also exhibits a relatively high positive error, suggesting the model predicts higher drag than expected.

Deep Interdiction Aircraft: Detailed Analysis

Component-Level Breakdown at Mach 1.6:

1 Wing Contributions: - Cd0_wing: 0.017471 - Cl_alpha_wing: 0.01573 - Wing aspect ratio (1.27) creates higher induced drag compared to F-16 (AR=3)	2 Fuselage/Body Contributions: - Cd0_body (skin friction): 0.00302 - Cd0_body (wave drag): 0.00197 - Total Cd0_body: 0.00499 - Larger fuselage diameter (3.3m) increases skin friction area
3 Drag Buildup Analysis: - Cdanc : 0.00347 - CdN2 (nose drag): 0.02670 - CdA (aftbody drag): 0.00043 - Total Cd0: 0.02246	4 Performance Metrics: - Cl (Lift coefficient): 0.1365 (Code) vs 0.146 (Expected) → -6.51% underestimation - Cd (Total drag): 0.0348 (Code) vs 0.0315 (Expected) → +10.48% overestimation - L/D ratio: 3.917 (Code) vs 4.65 (Expected) → -15.76% underestimation - Oswald efficiency (e): 0.5221 (lower than F-16 due to lower aspect ratio)

Detailed Component Breakdown: Code Output

Total Aircraft Drag (Cd)

The total aircraft drag coefficient calculated by the code is **0.03486**. This value incorporates contributions from various components.

Fuselage Zero-Lift Drag (Cd0)

The total zero-lift drag for the fuselage is **0.00499**, comprised of two main elements:

- Body Wave Drag: 0.00197
- Body Skin Friction: 0.00302

Induced Drag Analysis

- $K_{Induced}$ (Induced Drag due to lift): 0.4800
- $K_{Viscous}$ (Viscous Drag due to lift): 0.0850
- K_{Total} (Combined drag due to lift): 0.6651

Geometric Parameters Calculated

These factors show how lift generation contributes to total drag at Mach 1.6.

- Mean Aerodynamic Chord (MAC): 13.648 m
- Delta (Supersonic leading edge parameter): 72.39°
- Leading edge radius (rle): 0.25

These parameters are critical for accurate supersonic flow analysis and directly influence drag characteristics.

Pressure Coefficient Analysis

- The code calculates pressure distributions across the fuselage
- Wave drag is derived from pressure coefficient integration
- Shock locations are determined by Mach number and geometry

Understanding these individual drag components is crucial for optimizing aircraft design and enhancing overall aerodynamic efficiency. The code provides a granular view essential for such analyses.